



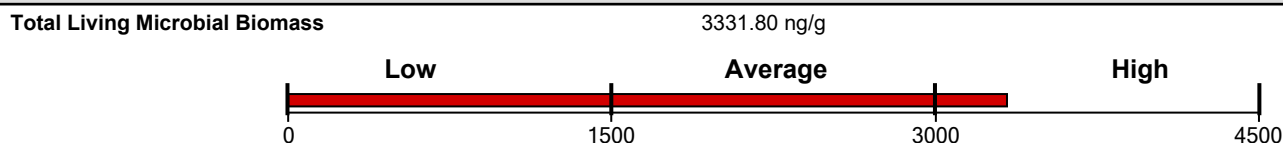
Account ID: 602314
MICHAEL LOBATO
1066 19 1/2 RD
FRUITA, CO 81521

Report Type: Biological
Invoice No.:
Date Received: 11/5/2024
Date Reported: 1/9/2025
Lab ID: B24-005478

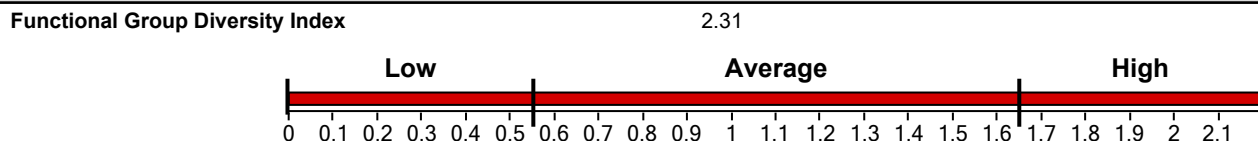
Results For: CSU WCRC FRUITA
Field ID: N-NOVEL BIOCHAR INJECTION
Sample ID: STRIP 1

PLFA Soil Microbial Community Analysis

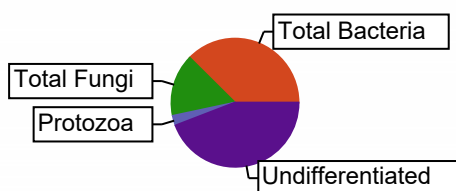
Functional Group Biomass



Functional Group Diversity



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	1249.11	37.49
Gram (+)	621.43	18.65
Actinomycetes	272.92	8.19
Gram (-)	627.68	18.84
Rhizobia	0.00	0.00
Total Fungi	520.11	15.61
Arbuscular Mycorrhizal	120.94	3.63
Saprophytes	399.16	11.98
Protozoa	82.53	2.48
Undifferentiated	1467.84	44.06



Community Composition Ratios

Fungi:Bacteria	0.42
Predator:Prey	0.07
Gram(+):Gram(-)	0.99

Stress & Community Activity Ratios

Saturated:Unsaturated	1.14
Monounsaturated:Polyunsaturated	3.92
Pre 16:1 w7c:17:0 cyclo	3.36
Pre 18:1 w7c:19:0 cyclo	None

Reviewed by: Wootton, Stephanie

All ratios should be looked at separately, but should also be taken into context and compared with one another to better understand the big picture. These are general guidelines and statements regarding soil microbial communities. In addition, the scales and ranges presented here are specific for the type of extraction and analytical methods used for PLFA analysis at Ward Laboratories, Inc. They will not necessarily reflect ranges derived from other methods of analysis or the literature. The scales can and should be adjusted slightly depending on the time of year and conditions at sampling along with the climate and soil type of specific regions where comparisons are being made. Conditions such as time of year, past and present crop, moisture, pH, and fertility should be noted or measured close to sampling for PLFA analysis for a more in depth interpretation of results.

Community Composition Ratios

Fungi : Bacteria

Bacteria tend to dominate in systems with fewer organic inputs or residues possibly leading to a lower C:N ratio. In addition, bacteria can be more prominent in the early spring or late fall as soil temperatures are usually cooler and vegetation is less active or absent. Dry conditions, slightly alkaline to alkaline pH values, or increased land disturbance through prolonged and extensive tillage, grazing, or compaction may also favor bacteria. While bacteria are important and needed in the soil ecosystem, fungi are desired and more often considered indicators of good soil health. Increased use of cover crops and/or other organic inputs and less soil disturbance should help the soil support more fungi. Adjustments to pH may also be recommended in some more extreme circumstances.

Predator : Prey

This ratio is also expressed as protozoa to bacteria. Protozoa feed on bacteria which helps release nutrients, especially nitrogen. A higher ratio indicates an active community where base level nutrients are sufficient to support higher trophic levels or predators. However, this ratio will always be a relatively low number because the prey will greatly outnumber the predators.

Gram (+) : Gram (-)

Gram (+) bacteria typically dominate early in the growing season and/or following a fallow period. They also survive better under certain environmental conditions or stressors such as drought or extreme temperatures due to their ability to form spores. Therefore, it is common to see higher values when the community is coming out of dormancy or is stressed. These values will typically begin to approach those of a more balanced bacterial community as the soil conditions become more favorable throughout the growing season. A gram (-) dominated soil may be due to anaerobic conditions or other stressors such as pesticide application or heavy metal contamination.

Stress and Community Activity Ratios

Saturated : Unsaturated

Bacteria alter their membranes under various environmental conditions in order to maintain optimal fluidity for nutrient and waste transport into and out of the cell. Saturated fatty acids may reflect a better adapted community to current environmental conditions. Communities under stressed conditions will increase their proportion of unsaturated fatty acids. This will likely occur most often as a result of low soil moisture or drastic changes in temperature. In general, a higher number indicates a healthier and more stable community.

Mono : Poly

The ratio of monounsaturated to polyunsaturated fatty acids is used along with the sat:unsat ratio to further indicate the degree of community stress. A higher ratio indicates less stress, while a lower ratio would depict higher levels of prolonged stress due to conditions such as temperature, moisture, pH, or nutrient availability (starvation).

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

Cyclo (cy) fatty acids are more prominent during stationary phases of growth or under high stress conditions that influence membrane fluidity and growth rates such as temperature, pH, moisture, and nutrient availability. In general, a higher number or all Pre16/Pre18 is better and indicates an actively growing community experiencing fewer stressors. These values are typically higher early in the growing season (planting) when the community is becoming active and experiencing fast growth. The values may begin to drop towards the end of the growing season (harvest) following a decrease in plant growth activity or as the community approaches a stationary growth phase as the temperature/moisture changes between the seasons.

Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

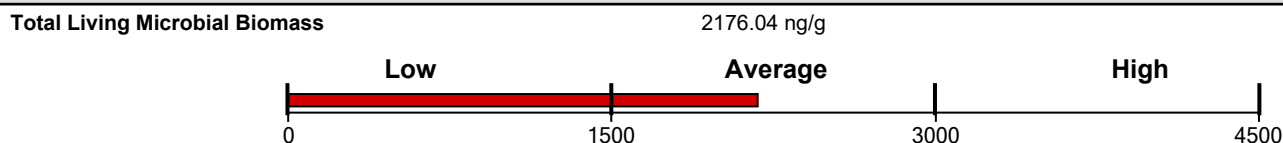
Account ID: 602314
MICHAEL LOBATO
1066 19 1/2 RD
FRUITA, CO 81521

Report Type: Biological
Invoice No.:
Date Received: 11/5/2024
Date Reported: 1/9/2025
Lab ID: B24-005479

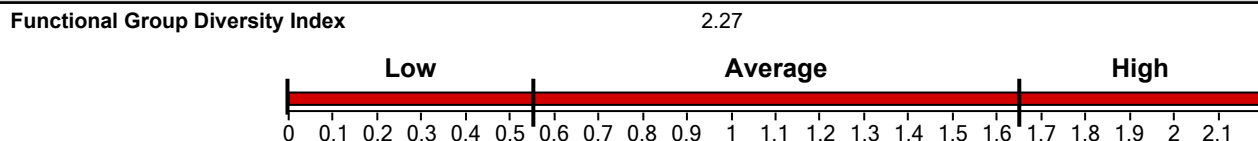
Results For: CSU WCRC FRUITA
Field ID: N-NOVEL BIOCHAR INJECTION
Sample ID: STRIP 2

PLFA Soil Microbial Community Analysis

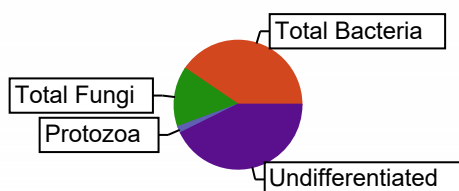
Functional Group Biomass



Functional Group Diversity



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	879.55	40.42
Gram (+)	370.13	17.01
Actinomycetes	165.76	7.62
Gram (-)	509.42	23.41
Rhizobia	0.00	0.00
Total Fungi	326.11	14.99
Arbuscular Mycorrhizal	80.87	3.72
Saprophytes	245.24	11.27
Protozoa	35.94	1.65
Undifferentiated	926.63	42.58



Community Composition Ratios

Fungi:Bacteria	0.37
Predator:Prey	0.04
Gram(+):Gram(-)	0.73

Stress & Community Activity Ratios

Saturated:Unsaturated	1.23
Monounsaturated:Polyunsaturated	3.73
Pre 16:1 w7c:17:0 cyclo	3.92
Pre 18:1 w7c:19:0 cyclo	2.68

Reviewed by: Wootton, Stephanie

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Community Composition Ratios

Fungi : Bacteria

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Gram (+) : Gram (-)

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Stress and Community Activity Ratios

Saturated : Unsaturated

Bacteria alter their membranes under various environmental conditions in order to maintain optimal fluidity for nutrient and waste transport into and out of the cell. Saturated fatty acids may reflect a better adapted community to current environmental conditions. Communities under stressed conditions will increase their proportion of unsaturated fatty acids. This will likely occur most often as a result of low soil moisture or drastic changes in temperature. In general, a higher number indicates a healthier and more stable community.

Mono : Poly

The ratio of monounsaturated to polyunsaturated fatty acids is used along with the sat:unsat ratio to further indicate the degree of community stress. A higher ratio indicates less stress, while a lower ratio would depict higher levels of prolonged stress due to conditions such as temperature, moisture, pH, or nutrient availability (starvation).

Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

Cyclo (cy) fatty acids are more prominent during stationary phases of growth or under high stress conditions that influence membrane fluidity and growth rates such as temperature, pH, moisture, and nutrient availability. In general, a higher number or all Pre16/Pre18 is better and indicates an actively growing community experiencing fewer stressors. These values are typically higher early in the growing season (planting) when the community is becoming active and experiencing fast growth. The values may begin to drop towards the end of the growing season (harvest) following a decrease in plant growth activity or as the community approaches a stationary growth phase as the temperature/moisture changes between the seasons.

Diversity Index Ratings

Total Biomass	Diversity	Rating
< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated

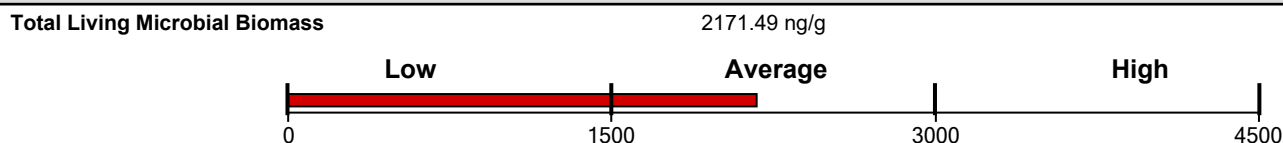
Account ID: 602314
 MICHAEL LOBATO
 1066 19 1/2 RD
 FRUITA, CO 81521

Report Type: Biological
 Invoice No.:
 Date Received: 11/5/2024
 Date Reported: 1/9/2025
 Lab ID: B24-005480

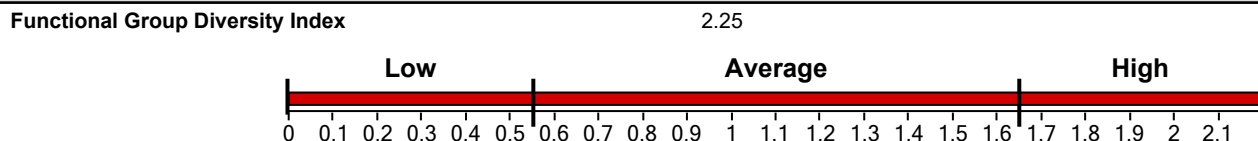
Results For: CSU WCRC FRUITA
 Field ID: N-NOVEL BIOCHAR INJECTION
 Sample ID: STRIP 3

PLFA Soil Microbial Community Analysis

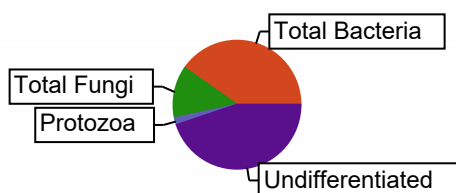
Functional Group Biomass



Functional Group Diversity



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	871.86	40.15
Gram (+)	382.81	17.63
Actinomycetes	191.20	8.80
Gram (-)	489.05	22.52
Rhizobia	0.00	0.00
Total Fungi	282.08	12.99
Arbuscular Mycorrhizal	95.11	4.38
Saprophytes	186.97	8.61
Protozoa	38.44	1.77
Undifferentiated	970.49	44.69



Community Composition Ratios

Fungi:Bacteria	0.32
Predator:Prey	0.04
Gram(+):Gram(-)	0.78

Stress & Community Activity Ratios

Saturated:Unsaturated	1.47
Monounsaturated:Polyunsaturated	5.25
Pre 16:1 w7c:17:0 cyclo	3.92
Pre 18:1 w7c:19:0 cyclo	2.07

Reviewed by: Wootton, Stephanie

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Community Composition Ratios

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Stress and Community Activity Ratios

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Mono : Poly

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Pre 16:1w7c : cy17:0 - Pre 18:1w7c : cy19:0

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Diversity Index Ratings

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< 500	< 1.0	Very Poor
500+ -> 0.1	1.0+ -> 1.1	Poor
1000+ -> 1500	1.1+ -> 1.2	Slightly Below Average
1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
0.05+ -> 0.1	Poor
0.1+ -> 0.15	Slightly Below Average
0.15+ -> 0.2	Average
0.2+ -> 0.25	Slightly Above Average
0.25+ -> 0.3	Good
0.3+ -> 0.35	Very Good
> 0.35	Excellent

Predator:Prey Ratings

Scale	Rating
< 0.002	Very Poor
0.002+ -> 0.005	Poor
0.005+ -> 0.008	Slightly Below Average
0.008+ -> 0.01	Average
0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated



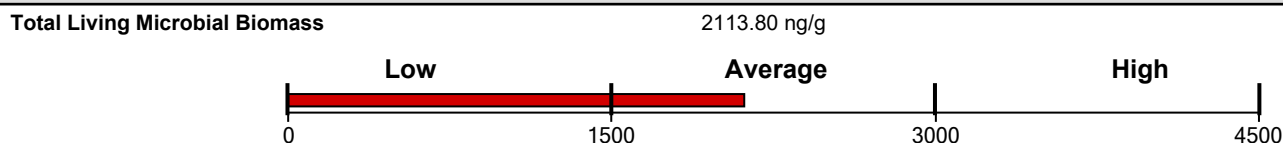
Account ID: 602314
MICHAEL LOBATO
1066 19 1/2 RD
FRUITA, CO 81521

Report Type: Biological
Invoice No.:
Date Received: 11/5/2024
Date Reported: 1/9/2025
Lab ID: B24-005481

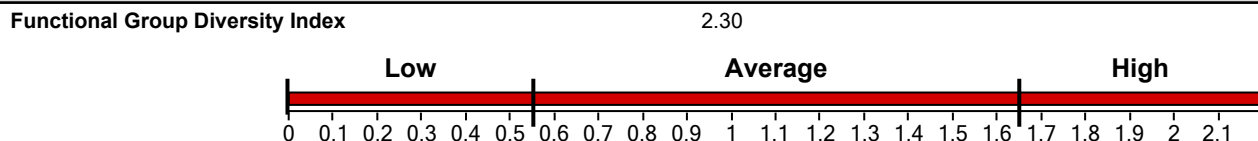
Results For: CSU WCRC FRUITA
Field ID: N-NOVEL BIOCHAR INJECTION
Sample ID: STRIP 4

PLFA Soil Microbial Community Analysis

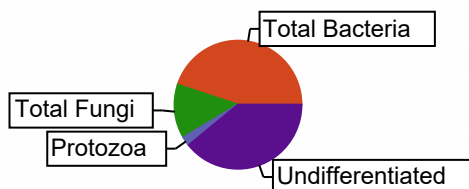
Functional Group Biomass



Functional Group Diversity



Functional Group	Biomass, ng/g	Percent, %
Total Bacteria	946.34	44.77
Gram (+)	394.81	18.68
Actinomycetes	193.33	9.15
Gram (-)	551.54	26.09
Rhizobia	0.00	0.00
Total Fungi	286.78	13.57
Arbuscular Mycorrhizal	87.90	4.16
Saprophytes	198.87	9.41
Protozoa	48.60	2.30
Undifferentiated	823.86	38.98



Community Composition Ratios

Fungi:Bacteria	0.30
Predator:Prey	0.05
Gram(+):Gram(-)	0.72

Stress & Community Activity Ratios

Saturated:Unsaturated	1.45
Monounsaturated:Polyunsaturated	6.10
Pre 16:1 w7c:17:0 cyclo	3.63
Pre 18:1 w7c:19:0 cyclo	2.06

Reviewed by: Wootton, Stephanie

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1500+ -> 2500	1.2+ -> 1.3	Average
2500+ -> 3000	1.3+ -> 1.4	Slightly Above Average
3000+ -> 3500	1.4+ -> 1.5	Good
3500+ -> 4000	1.5+ -> 1.6	Very Good
> 4000	> 1.6	Excellent

Fungi:Bacteria Ratings

Scale	Rating
< 0.05	Very Poor
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0.1+ -> 0.15	Slightly Below Average
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0.3+ -> 0.35	Very Good
> 0.35	Excellent

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Scale	Rating
< 0.002	Very Poor
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0.01+ -> 0.013	Slightly Above Average
0.013+ -> 0.016	Good
0.016+ -> 0.02	Very Good
> 0.02	Excellent

Gram (+):Gram (-) Ratings

Scale	Rating
< 0.5	Gram (-) Dominated
0.5+ -> 1	Slightly Gram (-) Dominated
1+ -> 2	Balanced Bacterial Community
2+ -> 3	Slightly Gram(+) Dominated
3+ -> 4	Gram(+) Dominated
> 4	Very Gram(+) Dominated